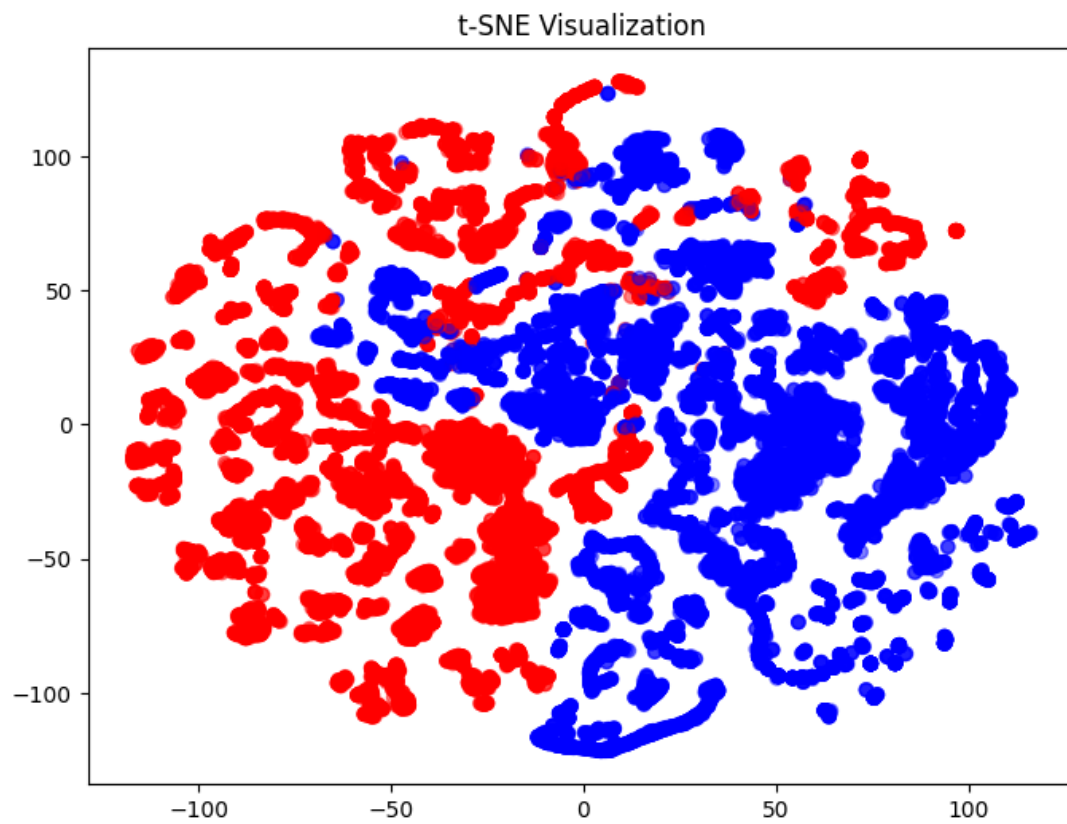


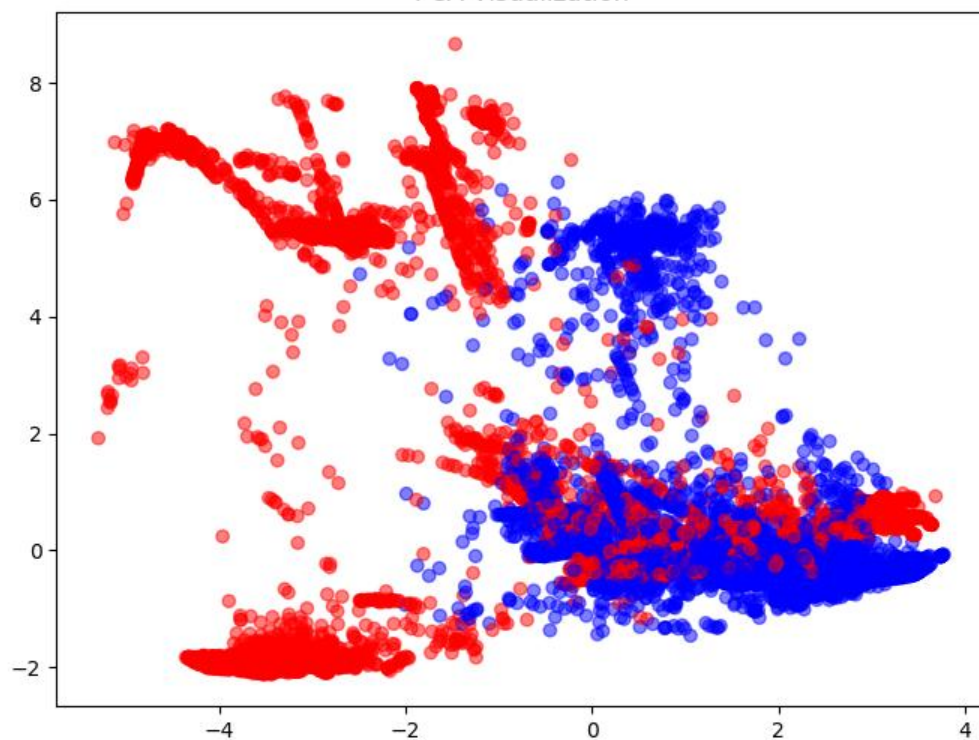
EE446_Assignment_2

Date: 8/14/26

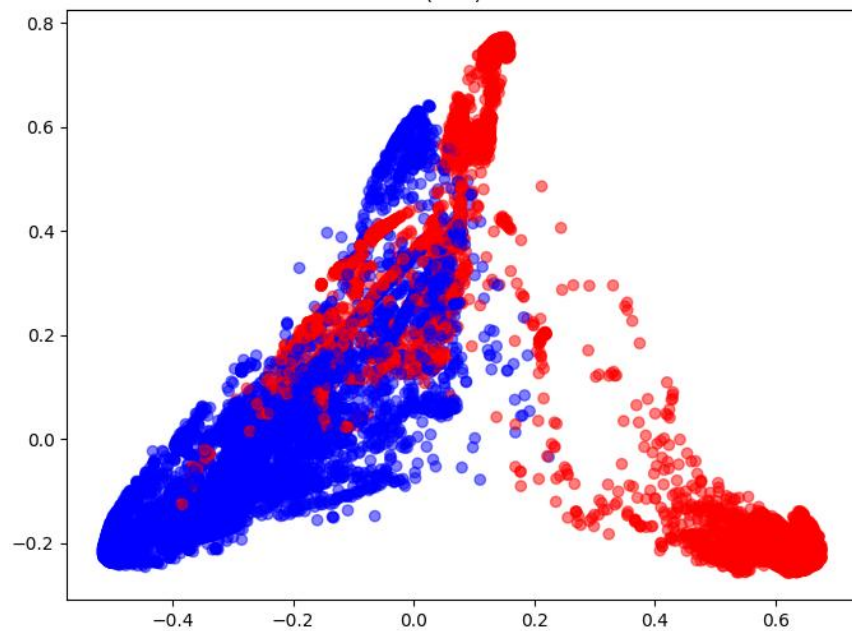
Plotting of the graphs:



PCA Visualization



Kernel PCA (RBF) Visualization



Classification and matrix writeup report:

```
y_pred = (y_pred_prob > 0.5).astype(int)
print("original model classification report: ")
print(classification_report(y_test, y_pred))
print (" original model confusion matrix:")
print(confusion_matrix(y_test, y_pred))
```

5] ✓ 1.1s Python

788/788 [=====] - 1s 829us/step

original model classification report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	11749
1	1.00	1.00	1.00	13446
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

original model confusion matrix:

```
[[11738  11]
 [  28 13418]]
```

Quantized classification and confusion matrix report:

```
print("Quantized Model Confusion Matrix:")  
print(confusion_matrix(y_test, y_pred_quant))
```

9]

✓ 0.5s

Python

Quantized Model Classification Report:

	precision	recall	f1-score	support
0	0.47	1.00	0.64	11749
1	0.00	0.00	0.00	13446
accuracy			0.47	25195
macro avg	0.23	0.50	0.32	25195
weighted avg	0.22	0.47	0.30	25195

Quantized Model Confusion Matrix:

```
[[11749    0]  
 [13446    0]]
```

Serial Monitor output from the five samples of the initial C++ code:

```
146         X_test[i][j],
147         input->params.scale,
148         input->params.zero_point
149     );
150 }
151
152 // Run the model on this input and check for error
153 if (interpreter->Invoke() != kTfLiteOk) {
154     Serial.println("Failed to invoke!");
155     continue;
156 }
157
158 // Question 5: Deploying the Quantized Model
159 // (a) Implement code to obtain the sigmoid prediction from the INT8
160 // output tensor and determine the predicted class label.
161 int8_t y_pred_quant = output->data.int8[0];
162 float prediction = (y_pred_quant - output->params.zero_point) * output->params.scale;
163 int predicted_class = (prediction > 0.5) ? 1 : 0;
164
165 // (b) Implement code to output Sample #, Predicted Class, and Actual Class for each sa
166 // INSERT YOUR CODE HERE //
167 Serial.print("Sample #");
```

Output Serial Monitor X

Not connected. Select a board and a port to connect automatically.

New Line 115200 baud

```
23:34:26.994 -> Sample #1, Predicted Class: 1, Actual Class: 1
23:34:28.131 -> Sample #2, Predicted Class: 1, Actual Class: 1
23:34:29.087 -> Sample #3, Predicted Class: 0, Actual Class: 0
23:34:30.117 -> Sample #4, Predicted Class: 0, Actual Class: 0
23:34:31.118 -> Sample #5, Predicted Class: 1, Actual Class: 1
```

The ten samples features and report on the serial monitor screenshot:

✓ → ⏏

🔌 Arduino Nano 33 BLE

📶

network_data.ino

network_model.h

```
200     if (input->type != kTfLiteInt8 || output->type != kTfLiteInt8) {
201         Serial.println("Expected INT8 input and output tensors.");
202         return;
203     }
204 }
205
206 void loop() {
207     for (uint8_t i = 0; i < 10; i++) {
208         // Load the i-th test sample data into the input tensor
209         for (int j = 0; j < NUMBER_OF_INPUTS; j++) {
210             input->data.int8[j] = quantize_to_int8(
211                 X_test[i][j],
212                 input->params.scale,
213                 input->params.zero_point
214             );
215         }
216
217         // Run the model on this input and check for error
218         if (interpreter->Invoke() != kTfLiteOk) {
```

Output

Serial Monitor

✕

⌵ 🔊 ⋮

Not connected. Select a board and a port to connect automatically.

New Line ▾ 115200 baud

```
00:22:24.913 -> Sample #1, Predicted Class: 0, Actual Class: 0
00:22:25.852 -> Sample #2, Predicted Class: 0, Actual Class: 0
00:22:26.858 -> Sample #3, Predicted Class: 1, Actual Class: 1
00:22:27.881 -> Sample #4, Predicted Class: 1, Actual Class: 1
00:22:28.868 -> Sample #5, Predicted Class: 1, Actual Class: 1
00:22:29.868 -> Sample #6, Predicted Class: 1, Actual Class: 1
00:22:30.879 -> Sample #7, Predicted Class: 1, Actual Class: 1
00:22:31.865 -> Sample #8, Predicted Class: 1, Actual Class: 1
00:22:32.861 -> Sample #9, Predicted Class: 1, Actual Class: 1
00:22:33.893 -> Sample #10, Predicted Class: 0, Actual Class: 0
00:22:41.874 -> Sample #3, Predicted Class: 1, Actual Class: 1
00:22:43.735 -> Sample #4, Predicted Class: 1, Actual Class: 1
00:22:44.739 -> Sample #5, Predicted Class: 1, Actual Class: 1
00:22:45.731 -> Sample #6, Predicted Class: 1, Actual Class: 1
```

👤

Ln 207, Col 31 Arduino Nano 33 BLE on COM10 [not connected] 2